OXEOUS: MATHEMATICAL FOUNDATIONS & STATUTORY DERIVATIONS

Direct First-Principles Mathematical Invariants Resolving EUDR (Reg. (EU) 2023/1115 & Reg. (EU) 2025/2650) Compliance Deadlocks

EUR-Lex OJ L 150 Compliant | Zero Regulatory Hallucination | EIC Deep-Tech Moat

PILLAR 1 · SPATIAL & ONTOLOGICAL TENSOR ENGINE

EUR-Lex Art. 2(4), Art. 2(5)/2(6) & Recital 37

PWRI-3D Tri-Axial Tensor Adjudication

Industry Fatal Flaw: Conventional 2D GIS cannot distinguish primary forest clearing from cyclical cocoa, coffee, or oil palm renewal, triggering 66%–75% false positives across tropical smallholder farms (Thünen Institute, 2025).

DERIVED INVARIANT

$$M(x,y) = L_{spatial}(x,y) \otimes L_{temporal}(x,y) \cap L_{statutory}(x,y)$$

$$C(x,y) = \Phi_{MMU>0.5ha}\left[\Delta S_{T0\to Tcurr}(x,y) \cap B_{2020}(x,y)\right] \setminus \left[E_{Art.2(5)}(x,y) \cup E_{Art.2(3)/2(7)}(x,y)\right]$$

Art. 3(a) Violation (Red)
Canopy loss > 0.5 ha inside statutory 2020 forest; quarantined from EU entry.

Art. 2(5)/2(6) Replanting (Amber)
Cyclical tree-crop renewal on pre-2020 agricultural plantation; certified lawful.

Art. 2(3)/2(7) Silviculture (Blue)
Sustainable timber harvest inside managed planted forest without conversion.

Art. 3(a) Deforestation-Free (Green)
Zero post-2020 statutory disturbance across multi-sensor consensus layers.

PILLAR 2 · TOPOLOGICAL SEGREGATION STUDIO

EUR-Lex Art. 9(1)(d) & COM(2026) 191 Protocol

Continuous Topological Plot Carve-Out & Conservation

Statutory Impasse: Under Art. 9(1)(d), any unsegregated deforestation within a declared polygon disqualifies the entire plot. An isolated 1 ha edge clearing on a 150 ha cooperative estate causes 100% commercial forfeiture.

DERIVED INVARIANT

$$\mathcal{P}_{export} = \mathcal{P}_{declared} \setminus \mathcal{V}_{Art.3(a)}, \quad \mathcal{V}_{Art.3(a)} = \{\,(x,y) \in \mathcal{P}_{decl} \mid C(x,y) = Violation\,\}$$

$$\text{Geodesic Area Identity: } A_{total} = (A_{clean} + A_{replant} + A_{silviculture}) + A_{violation} = A_{export} + A_{violation}\ (\pm 0.00\ ha)$$

Planar Boolean Excision
Shapely 2.0 difference operator (P_decl \ V_3(a)) excises non-compliant polygons.

Exact Geodesic Conservation
WGS84 ellipsoidal integration proves ±0.00 ha area conservation in TRACES NT.

Compliant Harvest Salvage
Retains clean & Art. 2(5) exempt sub-polygons (0.0% statutory deforestation).

RFC 7946 6-Decimal GeoJSON
Self-healing bowtie topology repair ready for EU Single Window submission.

PILLAR 3 · ANTI-FRAUD MASS-BALANCE GATE

EUR-Lex Art. 9(1)(b), Art. 10(2)(f), Art. 16(3) & Reg. 2025/2650

Biophysical Yield & Commodity Laundering Invariant

Polygon-Laundering Risk: Without mass-balance gating, unscrupulous traders carve out a 1 ha clean parcel (or borrow neighbor coordinates) and claim a 50-tonne harvest came from it, laundering illicit volume through customs.

DERIVED INVARIANT

$$Y_{implied} = M_{decl} / \iint_{\mathcal{P}_{export}} dA \le Y_{ceiling}(c, region) \iff M_{decl} \le A_{export} \cdot Y_c$$

$$\text{Laundering Gate: } \text{CERTIFIED (PASS) if } Y_{implied} \le Y_c \text{ ; QUARANTINED (FRAUD SHIELD) if } Y_{implied} > Y_c$$

Implied Biophysical Yield
Computes M_declared / A_export [kg/ha/yr] against verified exportable area.

FAO / CIRAD / ICRAF Ceilings
Empirical regional biological caps across all 7 EUDR Annex I commodities.

Anti-Laundering Fraud Shield
Blocks shipments inflated beyond photosynthetic carrying capacity pre-customs.

Point-Evasion Audit (< 4 ha)
Screens Art. 9(1)(d) point declarations against perimeter evasion anomalies.

PILLAR 4 · MULTI-AGENT CONSENSUS & PROVENANCE

EUR-Lex Art. 3(b)/(c), Art. 25(2)(a) & Art. 33 (TRACES NT)

Six-Agent Deterministic Supervisor & SHA-256 Merkle Root

Audit & Turnover Liability: Optical alerts fail under tropical cloud cover while unconstrained LLMs hallucinate legal verdicts, exposing first operators to administrative fines ≥ 4% of EU annual turnover and Art. 17 seizures.

DERIVED INVARIANT

$$\Phi_{consensus} = \left[\sum w_k \cdot c_k \cdot \sigma_k\right] / \left[\sum w_k \cdot c_k\right] \ge \tau_{statutory}$$

$$H_{root} = SHA256(\,H(\mathcal{P}_{export}) \parallel H(C_{PWRI}) \parallel H(Y_{implied}) \parallel H(\Phi_{consensus})\,)$$

Six Concurrent Domain Agents
Optical (A1), SAR (A2), Cadastre (A3), Topology (A4), Legal (A5), Yield (A6).

Deterministic Supervisor Gate
Weighted fusion, strict IUCN/Art. 2(5) overrides & zero-hallucination gating.

SHA-256 Merkle Provenance
Immutable per-pixel & canonical JSON digest retained for 5-year statutory audit.

TARIC Box 44 · Code C716
Unlocks EU Customs Single Window (CERTEX) Due Diligence Statement clearance.

THE OXEOUS UNIFIED REGULATORY INVARIANT THEOREM

$$\mathcal{P}_{export} = \mathcal{P}_{declared} \setminus \mathcal{V}_{Art.3(a)} \quad \text{subject to:} \quad \left[M_{declared} / Area(\mathcal{P}_{export})\right] \le Y_{max}(c, region) \ \wedge \ \Phi_{consensus} \ge \tau_{statutory}$$

Simultaneously guarantees 100% deforestation-free EU customs clearance, eliminates up to 75% agroforestry false positives, and enforces a strict biophysical ceiling against polygon laundering.

TRACES NT · CERTEX
TARIC Box 44 · Code C716